

Chemical space: what is it and how to conceptualize it

In the meantime, get the latest version of
<https://github.com/jhelaja/CHEM-006> and go to
Laplaza_sessions and set up the environment

R. Laplaza

Jun 2025

What is chemical space?

“The theoretical collection of all possible molecules and chemical compounds that could exist, including those that have already been discovered and those yet to be synthesized”

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“The theoretical collection of all possible molecules and chemical compounds that could exist, including those that have already been discovered and those yet to be synthesized”

Basically, every molecule that makes sense.

How large is chemical space?

Infinite.

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Enumeration of small druglike molecules leads to 10^{60} molecules.

For reference, there are about 10^{80} atoms in the observable universe.

Chemical (sub)spaces are huge due to combinatorial explosions.

Reymond, J.-L.; Awale, M. *ACS Chem Neurosci* **2012**, 3 (9), 649

Why do we care about chemical space?

*Dealing with large multidimensional spaces is not very easy for human brains.
Visualization and manipulation of chemical spaces is insightful.*

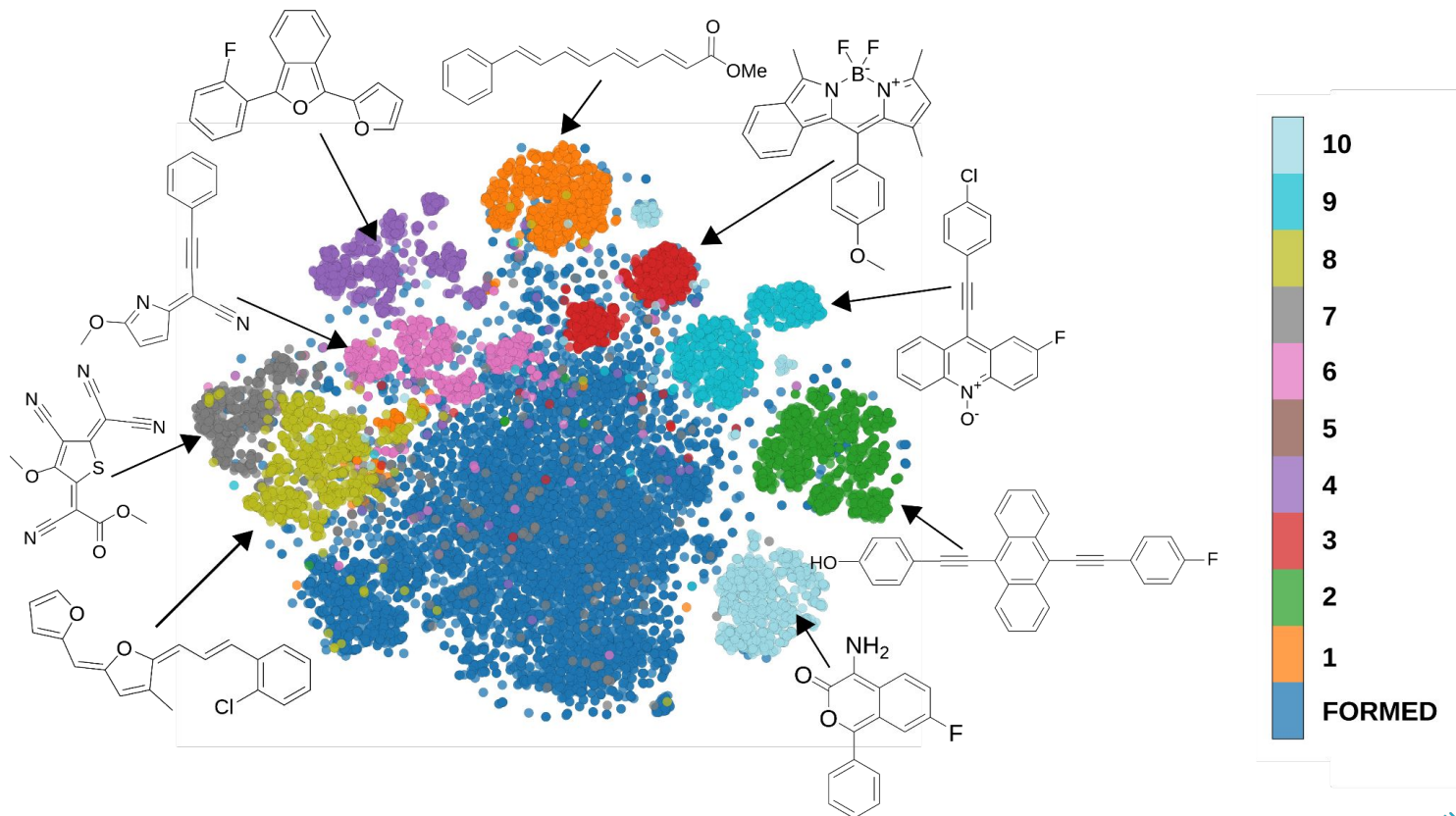
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Visualization and manipulation of chemical spaces is insightful.*

- Gives us estimates for screening, helps identify trends (in spite of cliffs)
- Gives us a feeling for diversity, lets us generate and enforce diversity.
- Lets us identify molecular neighbourhoods, families, retrosynthesis ideas.
- Lets us test applicability domains (e.g., scope of a reaction, ML extrapolation)

...

Why do we care about chemical space?



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There is no perfect way of visualizing/manipulating chemical space because there is **no universal definition of molecular similarity**.

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So, let's think about it as a set of tools more than an absolute truth.

Generative models and application in molecular design

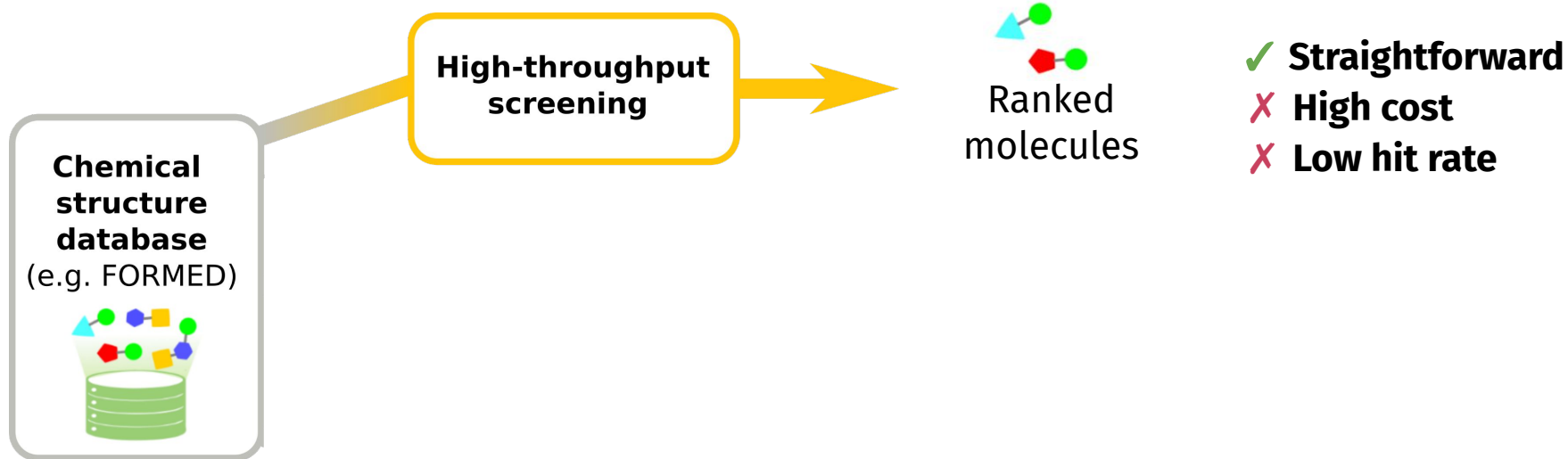
R. Laplaza
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What do we call a generative model in this context?

- We can train ML models to predict the properties of unseen molecules.
- However, this is not enough for most applications in chemistry, because **what we really really want is to find the best possible molecule**, i.e. the molecule that has the set of properties we want.
- This is what we basically spend our lives doing. Sometimes people differentiate between direct, rational and inverse design.
- In this context, a generative model is some piece of software that produces molecules when queried. *Note that this is a hand-wavy definition.*

Molecular design strategies

How to discover unknown molecules with tailored properties



HT(V)S

ML predictions allow us to screen millions of molecules really quickly

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PubChem (NIH + vendors) contains ~300 million molecules

Enamine REALSPACE contains ~70 billion molecules that can be made easily

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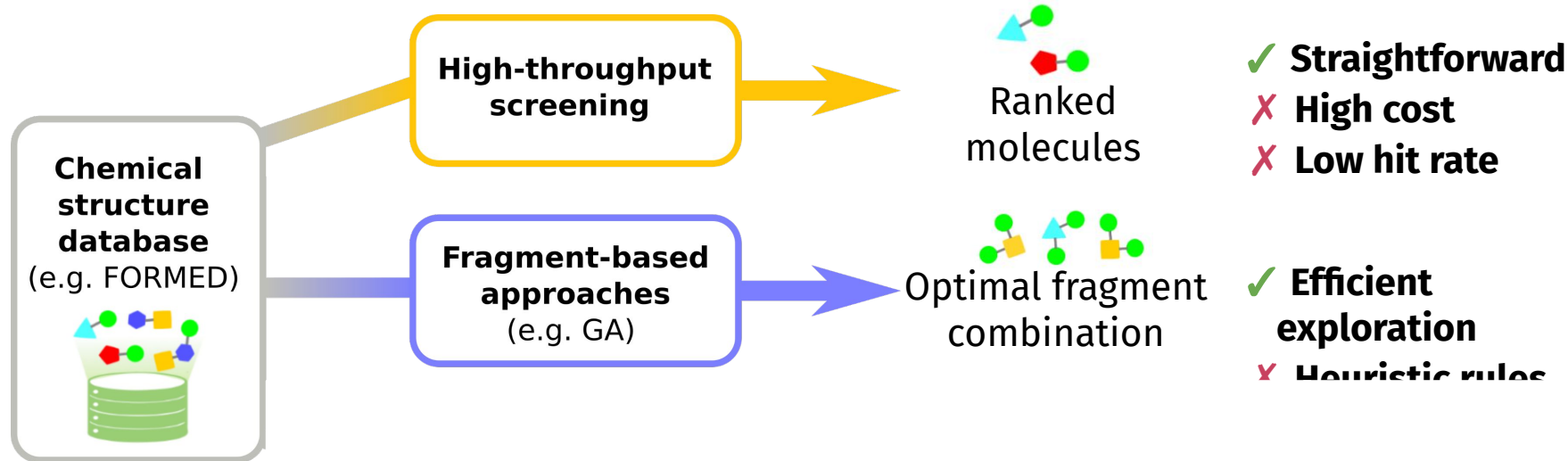
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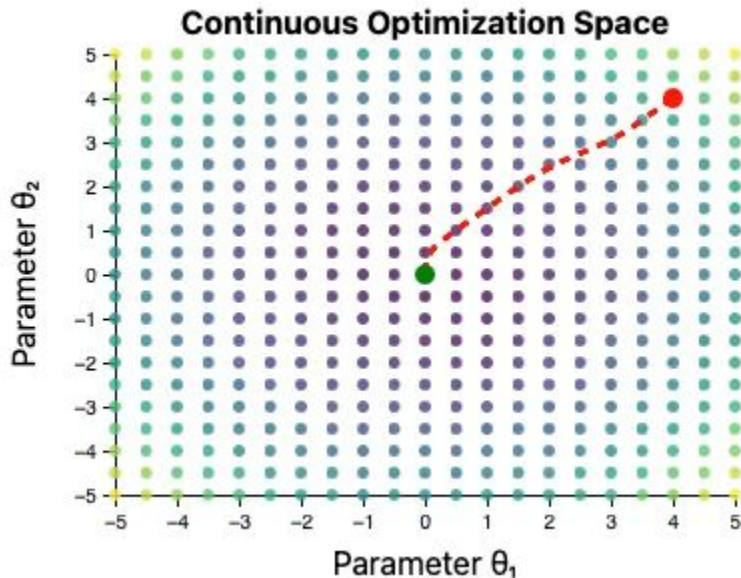
If it takes us one second to predict each molecule, that's ~**10** and ~**2000 years** for the first two, and like *until the heat death of the universe* for the third. *Oopsie!*

Molecular design strategies

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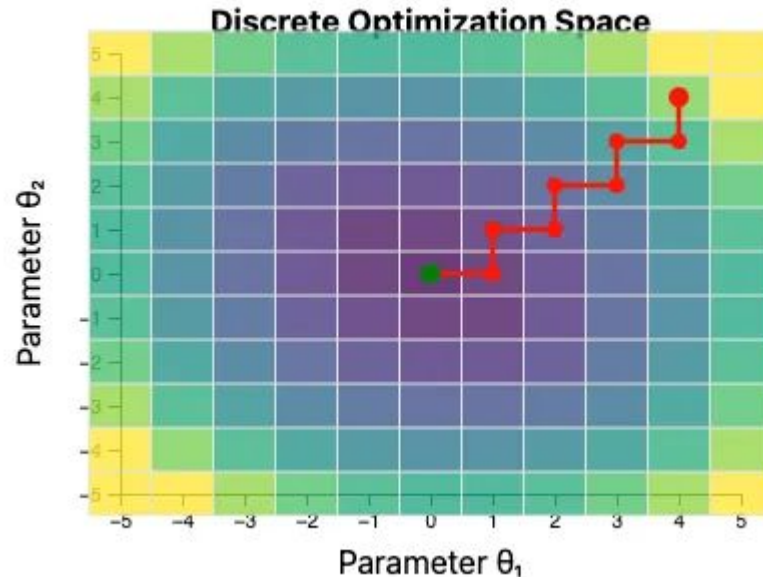
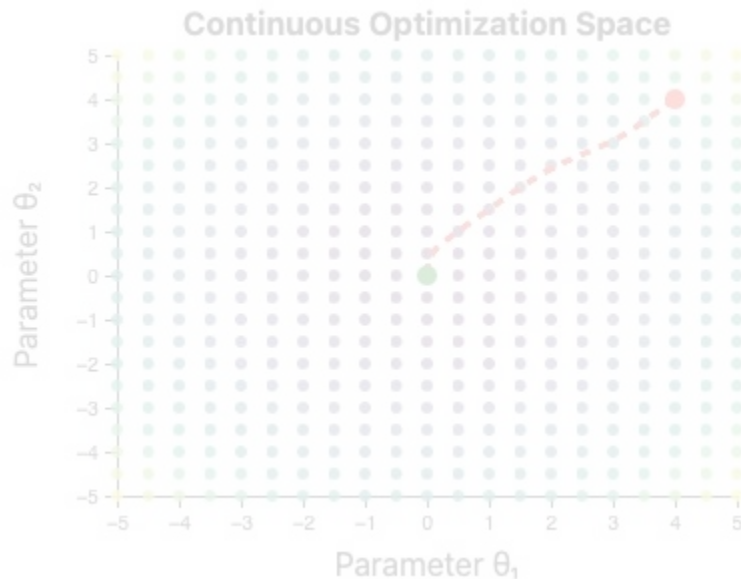
Optimization (of molecular structure/composition)



Let us imagine that we can project chemical space in two dimensions θ_1 and θ_2 and want to find molecules with good properties

From a starting point (**red**) we could compute the derivative of the property w.r.t. θ_1 and θ_2 , and go towards the closest molecule that improves the property until we reach the optimum (**green**)

Optimization (of molecular structure/composition)



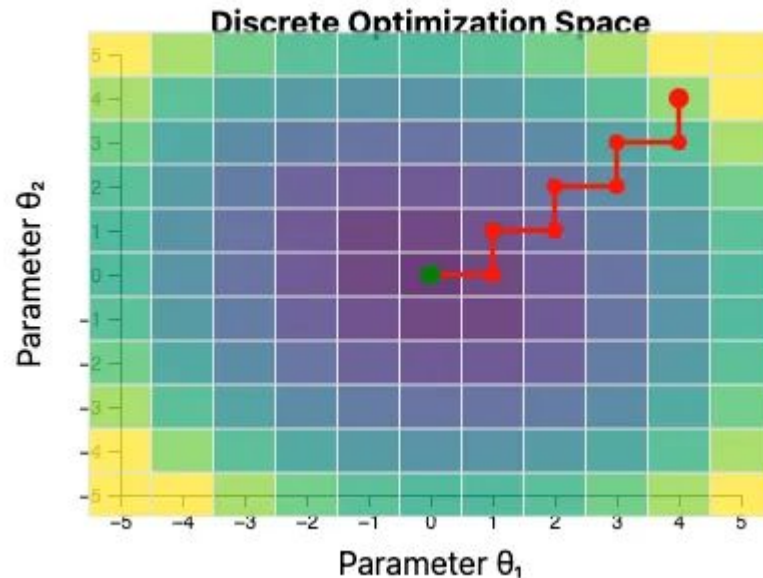
Optimization (of molecular structure/composition)

We cannot really define gradients in chemical structure!

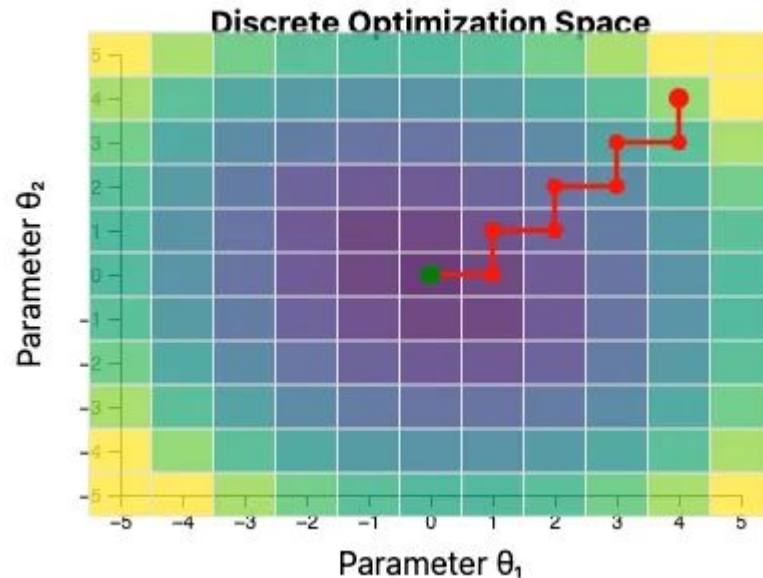
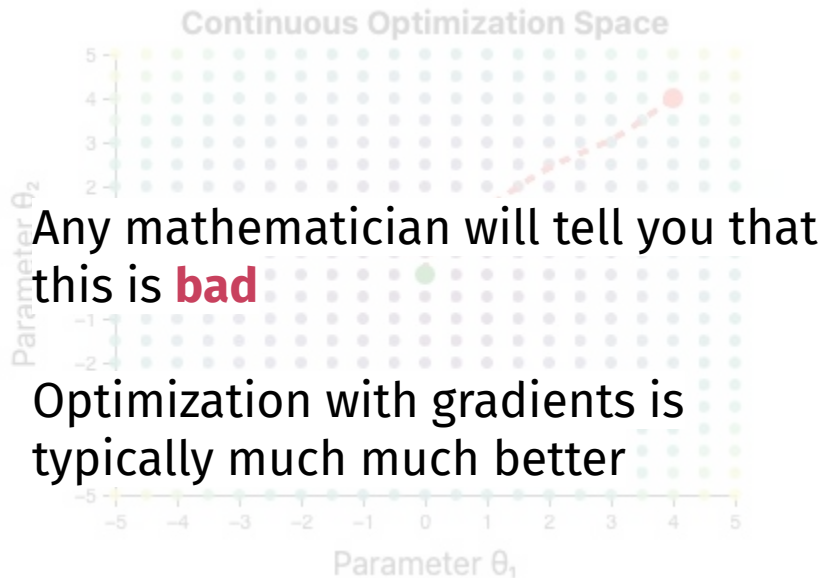
Derivatives ~ *infinitesimal change*

How do we change a C for a N “just a lil’ bit”??

This means *mathematically* our optimization is gradient-free



Optimization (of molecular structure/composition)

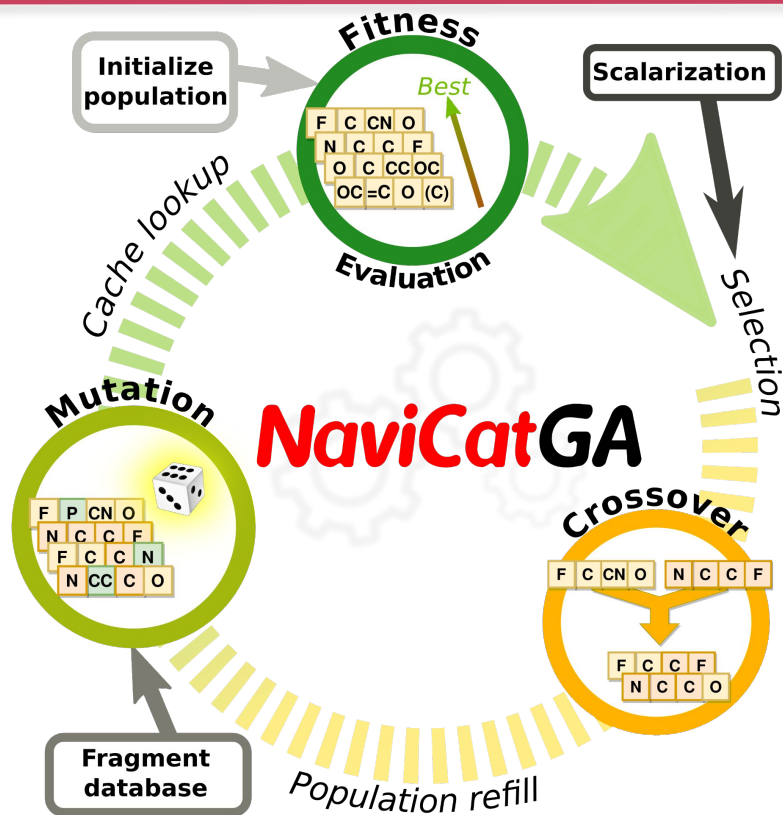


A pragmatic solution: GAs



This is the way to go if we want to get things done

The “natural” solution for molecular design is dealing with discrete things... discretely



Laplaza, R.; Gallarati, S.; Corminboeuf, C. *Chemistry Methods* **2022**, 2 (6), e202100107

Foscato, M.; Jensen, V. R. *ACS Catal.* **2020**, 10 (3), 2354

Extensive work by Jensen and Foscato (Bergen)

Seumer, J.; Kirschner Solberg Hansen, J.; Brøndsted Nielsen, M.; Jensen, J. H. *ANIE* **2023**, 135 (18), e202218565

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That is, no gradients are ever defined and we use gradient-free optimization*

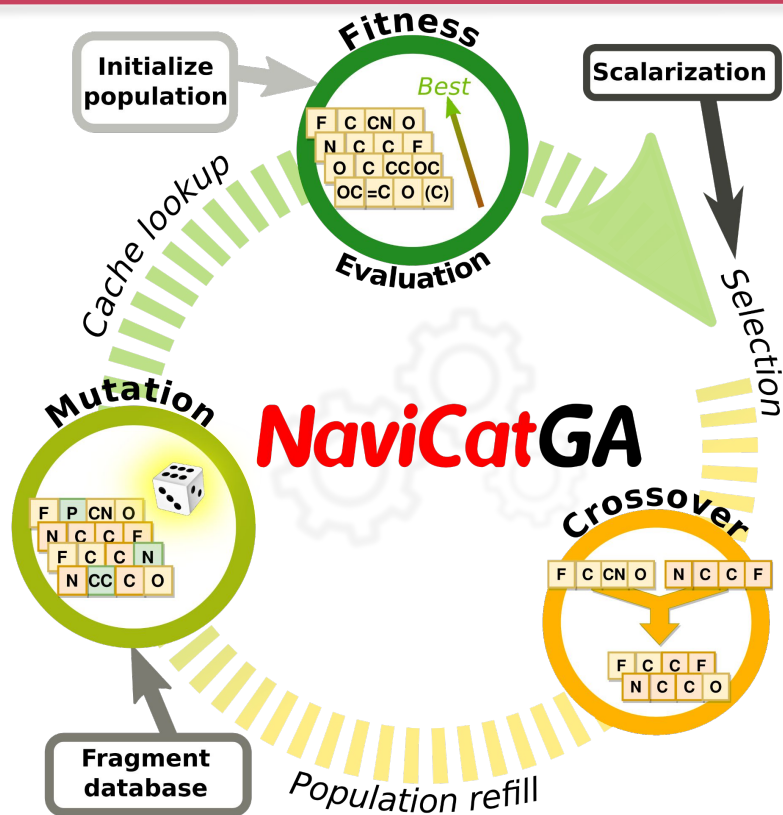
*GA only an example, there are other types (e.g. PSO) and many flavors of GA

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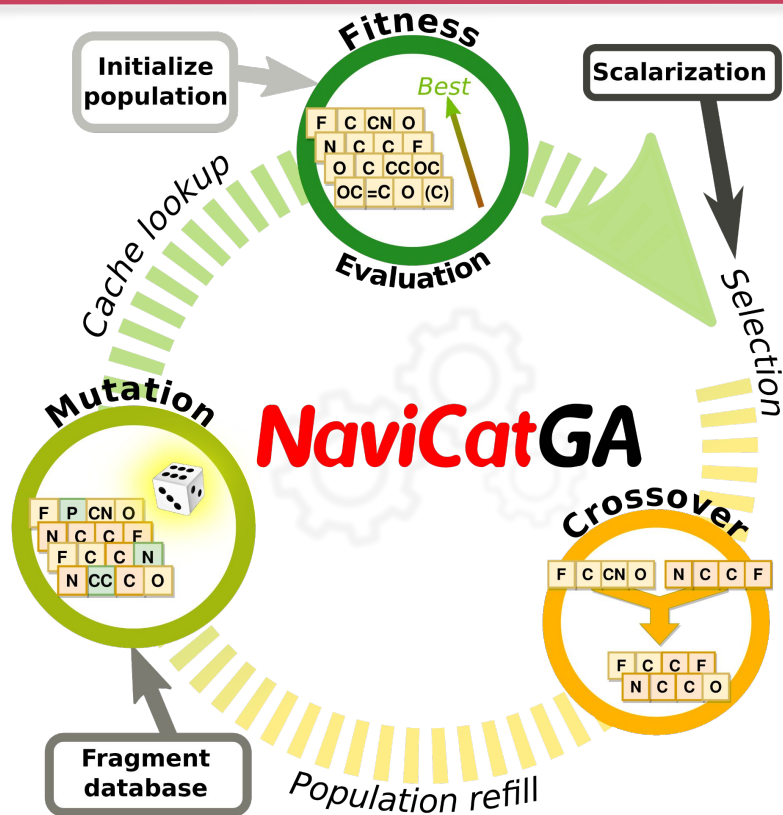


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Main advantage: no gradients → any model or models, at any point



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- Ease of combining with ML models
- Bottle-necked by speed of evaluation

Tripp, A.; Hernández-Lobato, J. M. Genetic Algorithms Are Strong Baselines for Molecule Generation. *arXiv* **2023**.

Method Source	REINVENT Gao et al. (2022)	Graph GA Gao et al. (2022)	MOL_GA Our experiments
albuterol_similarity	0.882± 0.006	0.838± 0.016	0.896±0.035
amlodipine_mpo	0.635± 0.035	0.661± 0.020	0.688±0.039
celecoxib_rediscovery	0.713± 0.067	0.630± 0.097	0.567±0.083
deco_hop	0.666± 0.044	0.619± 0.004	0.649±0.025
drd2	0.945± 0.007	0.964± 0.012	0.936±0.016
fexofenadine_mpo	0.784± 0.006	0.760± 0.011	0.825±0.019
gsk3b	0.865± 0.043	0.788± 0.070	0.843±0.039
isomers_c7h8n2o2	0.852± 0.036	0.862± 0.065	0.878±0.026
isomers_c9h10n2o2pf2cl	0.642± 0.054	0.719± 0.047	0.865±0.012
jnk3	0.783± 0.023	0.553± 0.136	0.702±0.123
median1	0.356± 0.009	0.294± 0.021	0.257±0.009
median2	0.276± 0.008	0.273± 0.009	0.301±0.021
mestranol_similarity	0.618± 0.048	0.579± 0.022	0.591±0.053
osimertinib_mpo	0.837± 0.009	0.831± 0.005	0.844±0.015
perindopril_mpo	0.537± 0.016	0.538± 0.009	0.547±0.022
qed	0.941± 0.000	0.940± 0.000	0.941±0.001
ranolazine_mpo	0.760± 0.009	0.728± 0.012	0.804±0.011
scaffold_hop	0.560± 0.019	0.517± 0.007	0.527±0.025
sitagliptin_mpo	0.021± 0.003	0.433± 0.075	0.582±0.040
thiothixene_rediscovery	0.534± 0.013	0.479± 0.025	0.519±0.041
troglitazone_rediscovery	0.441± 0.032	0.390± 0.016	0.427±0.031
valsartan_smarts	0.178± 0.358	0.000± 0.000	0.000±0.000
zaleplon_mpo	0.358± 0.062	0.346± 0.032	0.519 ± 0.029
Sum	14.196	13.751	14.708
Old Rank	1	2	N/A
New Rank	2	3	1

Fragment
database

Population refill

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A pragmatic solution: GAs

This is the way to

Initialize
population

Fitness

Best

Scalarization

than essential, especially if these baselines might perform well. We propose to address this by introducing the *GA criterion*: a rule of thumb that new methods in molecular generation should offer some advantage over GAs. This advantage can either be empirical (e.g. outperforming GAs in experiments) but also conceptual: what limitation of randomly changing known molecules does the proposed method potentially overcome? We call upon members of this community to enforce this standard in the peer review process.

- Our results also suggest another intriguing possibility: maybe newer algorithms do not outperform GAs because *they are also just generating variants of known molecules*, just in an indirect way. Indeed, many newer methods are essentially generative models which are trained on a dataset of known molecules. It would therefore not be surprising if the novel molecules generated by such methods are simply variants of their training data. We think investigating this is an important direction for future work.

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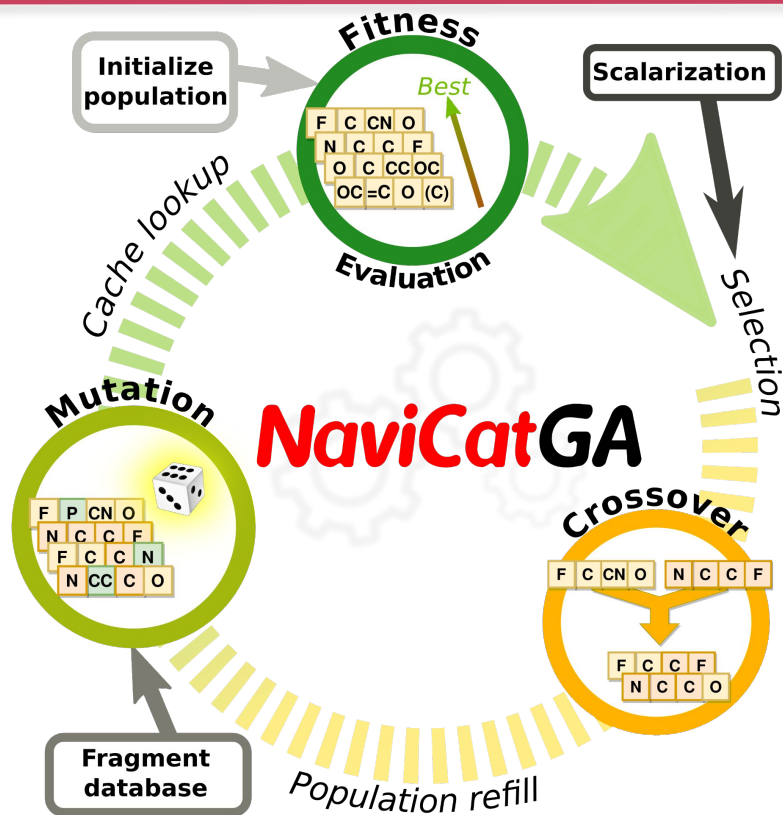
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This is the way to go if we want to get things done

Main disadvantage: heuristic rules and some degree of reliance on fragment databases



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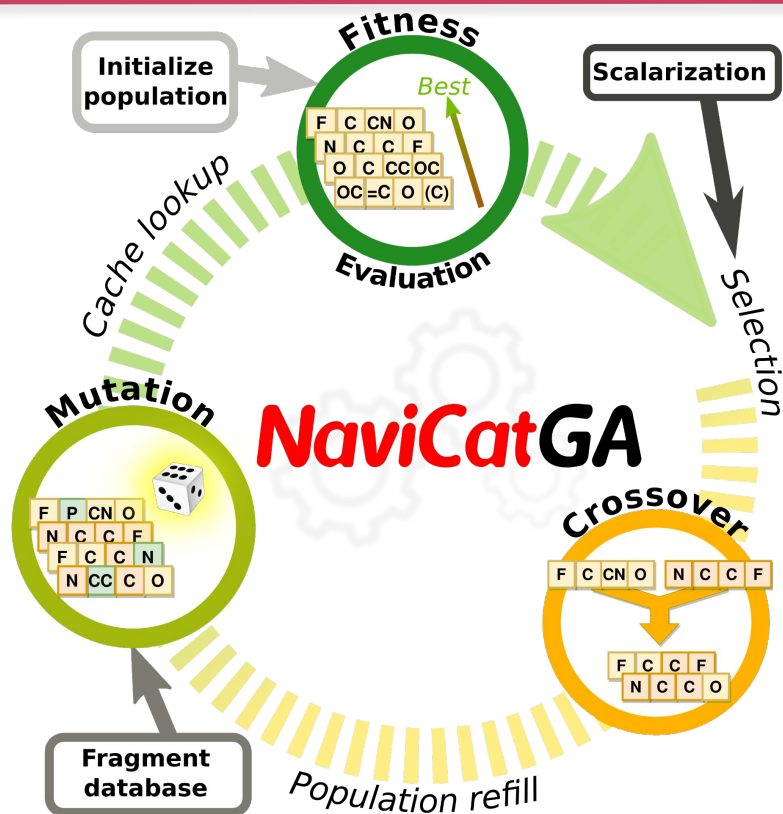
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- Defining a search space is often convenient and it's easy to reach billions of possibilities anyway



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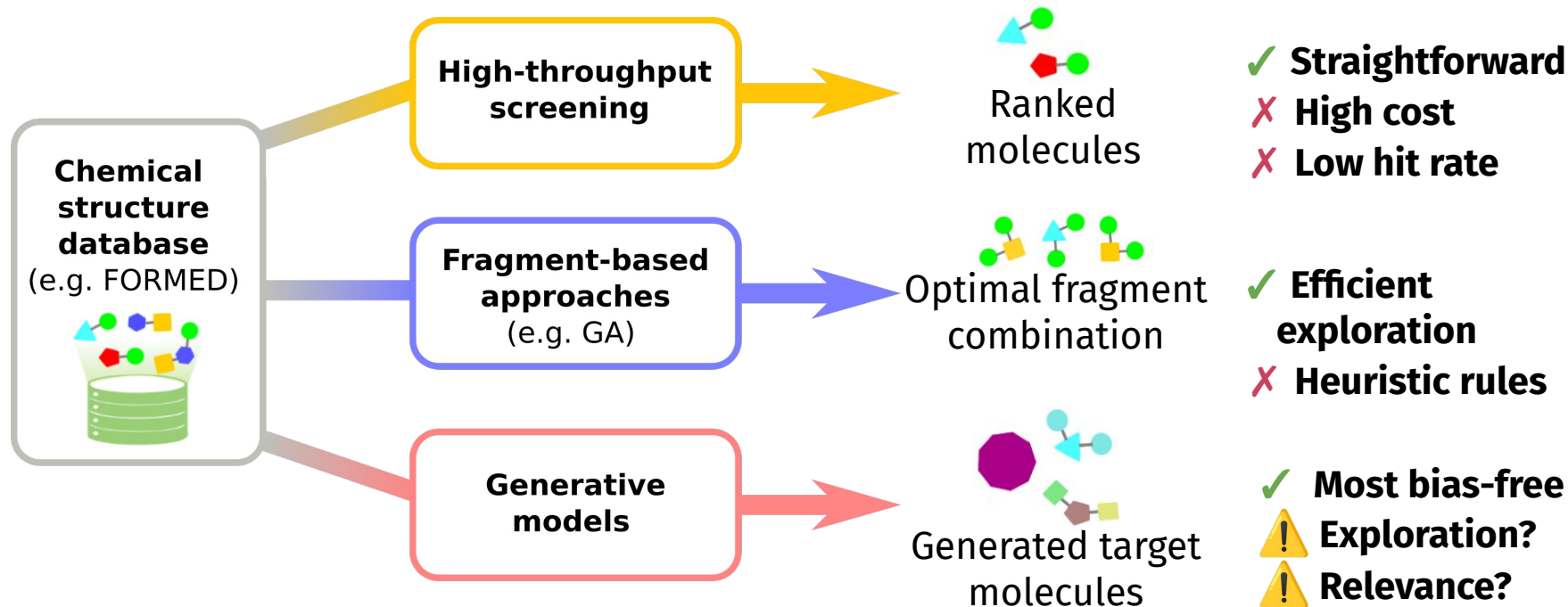
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Molecular design strategies

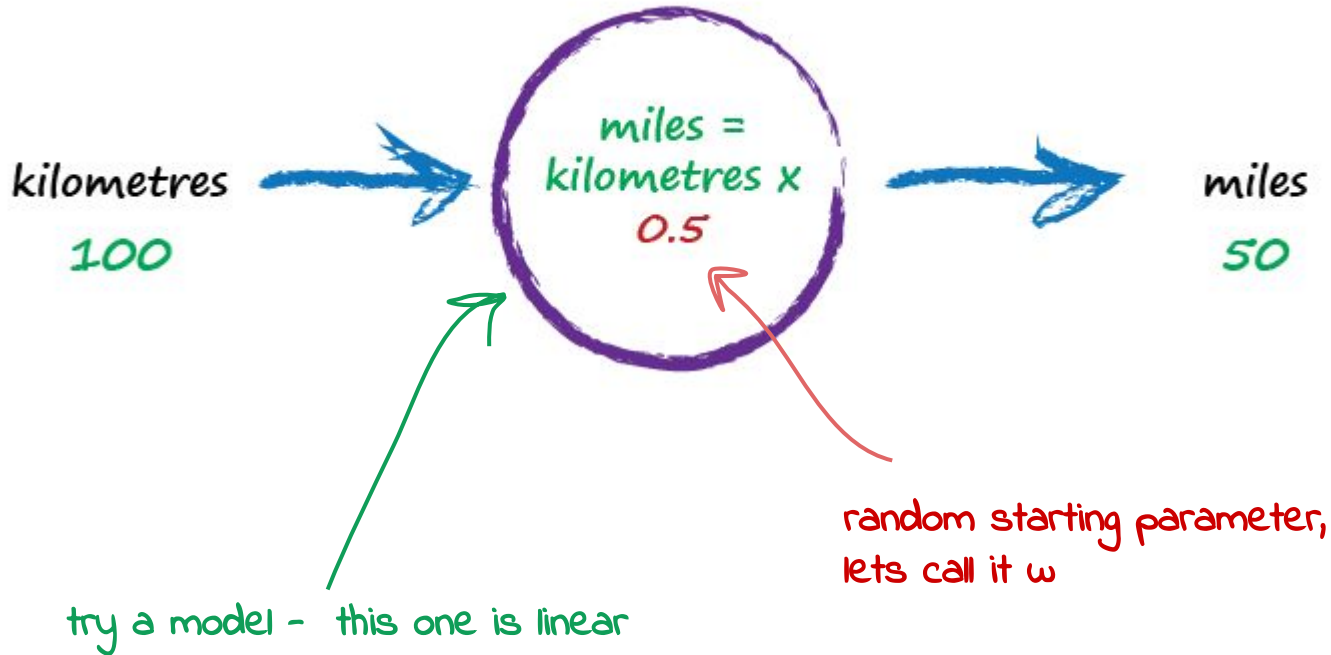
How to discover unknown molecules with tailored properties



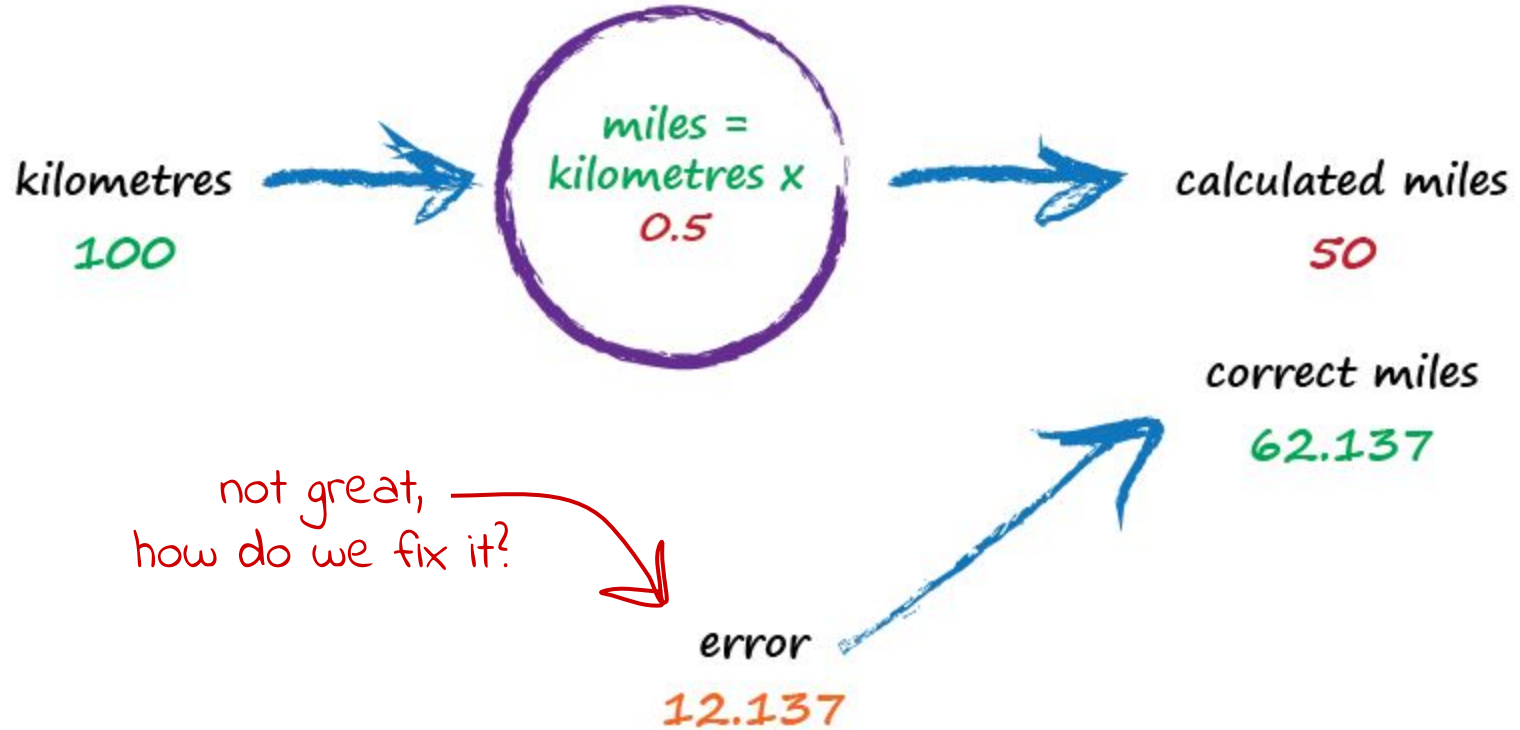
By the way: what the hell is a NN? My 2 cents

- NNs (virtually all architectures) are the love child of two technologies:
 - Automatic differentiation
 - Stochastic gradient descent optimization
- Example: we want to find the conversion between miles and km

Finding the conversion



Finding the conversion



Finding the conversion

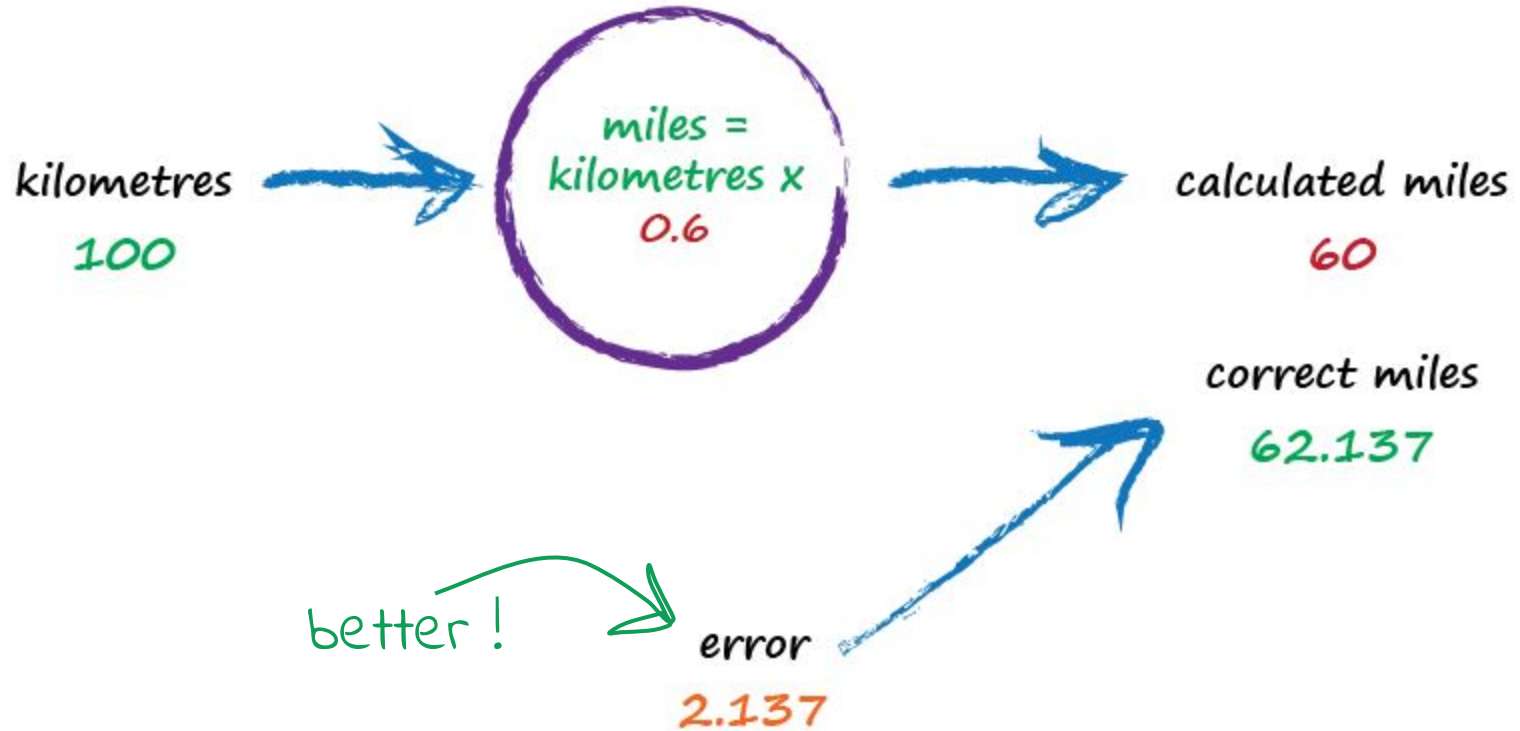
If we can compute the “gradient” dE/dw we can take a step in w in the direction to minimize the error E

We can define the error in many ways depending on the application!
Sometimes error is mean absolute error in a batch, sometimes its cross-entropy loss (for classification).

You can define your own error (loss function) *as long as it is differentiable!* In generative models, you normally define error as a *reconstruction loss*.

This is the core idea of training, nothing more, nothing less!

Finding the conversion



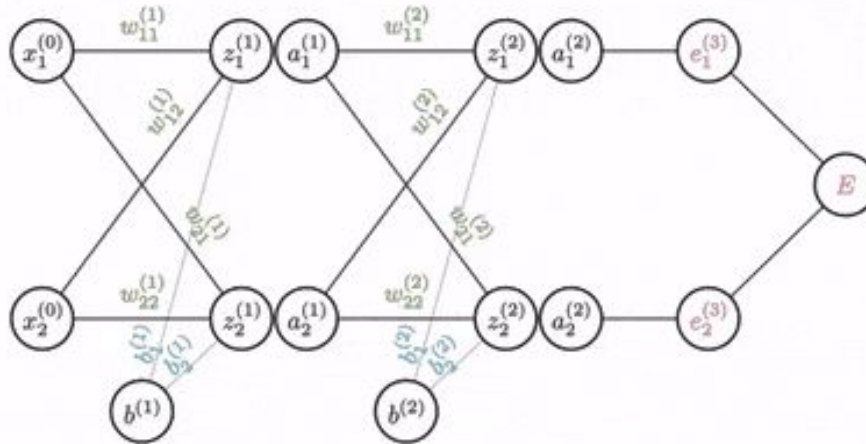
Finding the conversion

$$\frac{\partial E}{\partial w_{11}^{(2)}} = \frac{\partial e_1^{(3)}}{\partial a_1^{(2)}} \frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} \frac{\partial z_1^{(2)}}{\partial w_{11}^{(2)}}$$

$$\frac{\partial E}{\partial w_{12}^{(2)}} = \frac{\partial e_1^{(3)}}{\partial a_1^{(2)}} \frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} \frac{\partial z_1^{(2)}}{\partial w_{12}^{(2)}}$$

$$\frac{\partial E}{\partial w_{21}^{(2)}} = \frac{\partial e_2^{(3)}}{\partial a_2^{(2)}} \frac{\partial a_2^{(2)}}{\partial z_2^{(2)}} \frac{\partial z_2^{(2)}}{\partial w_{21}^{(2)}}$$

$$\frac{\partial E}{\partial w_{22}^{(2)}} = \frac{\partial e_2^{(3)}}{\partial a_2^{(2)}} \frac{\partial a_2^{(2)}}{\partial z_2^{(2)}} \frac{\partial z_2^{(2)}}{\partial w_{22}^{(2)}}$$



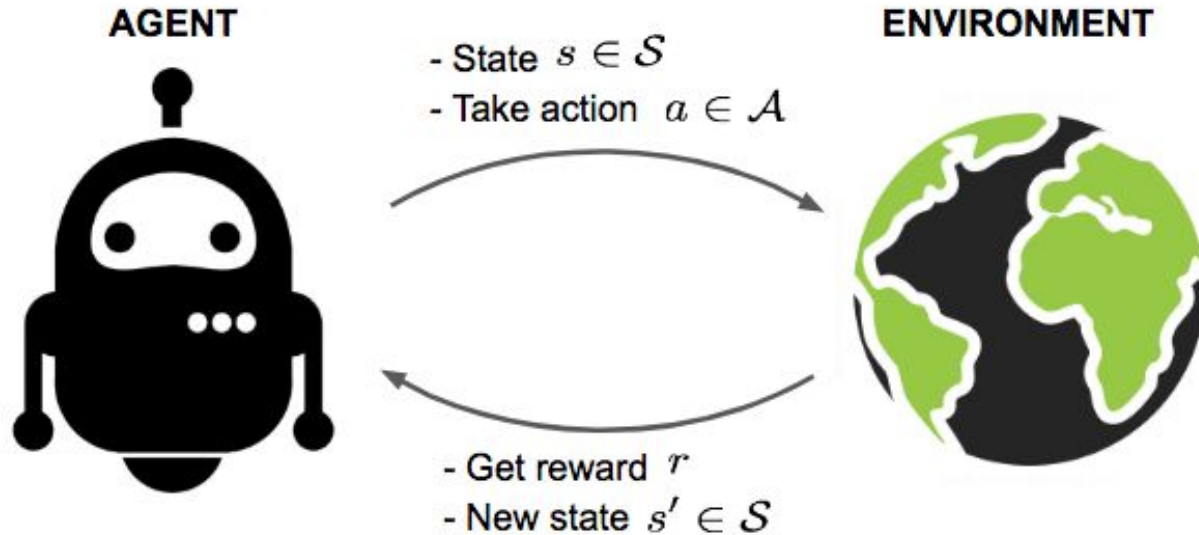
Training 101

- We have a bunch of layers or blocks with trainable weights whose output's gradient can be easily computed w.r.t. their weights.
- Then, the gradient of the output w.r.t. all the weights can be computed via chain rule.
- From the output and the reference data, we compute a loss function (error metric of choice), and we take the gradient of that.
- Then we adjust all the weights by a small step and iterate.

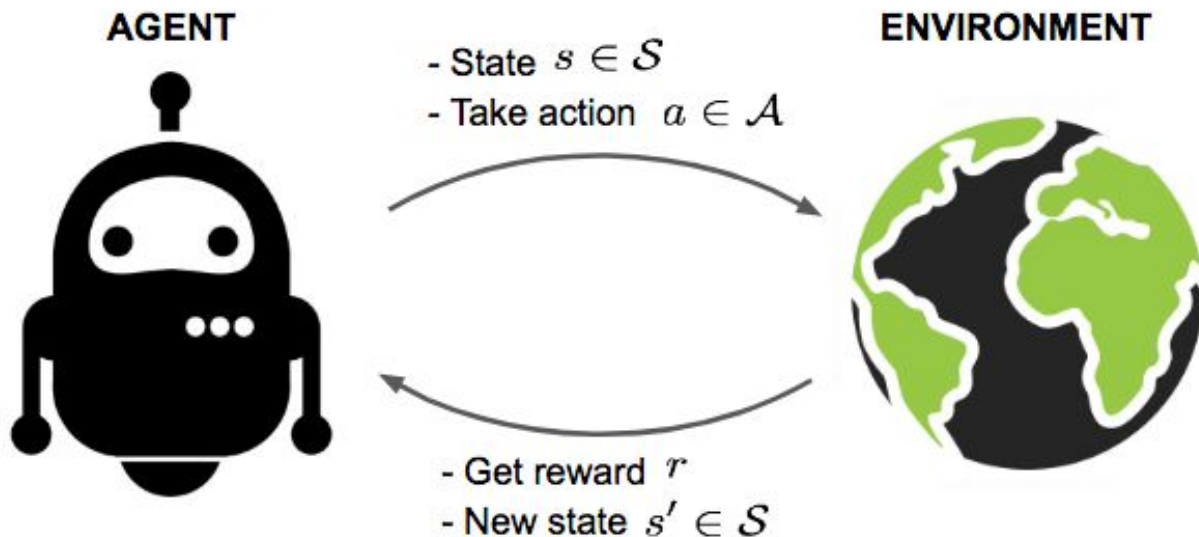
Back to topic! How do we generate molecules with ML?

- There are many ways with strong pros and cons. I cannot humanly cover what one would need to cover in the time we have.
- Instead, I'm gonna give a quick, opinionated overview.

The Reinforcement Learning approach



The Reinforcement Learning approach



The agent gets a reward and adjusts the probability of each action $\mathbf{a}$ (*policy*): $\pi(\mathbf{a}|\mathbf{s})$
A second loop can learn the *reward* surrogate $\mathbf{r}(\mathbf{s})$, which is called the *critic*.

State of the art of RL in 2020

Mastering Chess and Shogi by Self-Play with a General Reinforcement Learning Algorithm

David Silver,^{1*} Thomas Hubert,^{1*} Julian Schrittwieser,^{1*}
Ioannis Antonoglou,¹ Matthew Lai,¹ Arthur Guez,¹ Marc Lanctot,¹
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How do we do RL for board games?

$s \in S$: s is the board state, S all possible boards

$a \in A(s)$: legal moves given state s

$r(s')$: Reward function measures how good a move is by looking at new state s' after move

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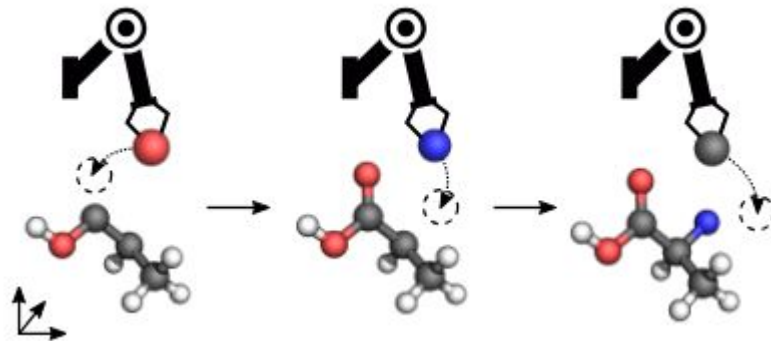
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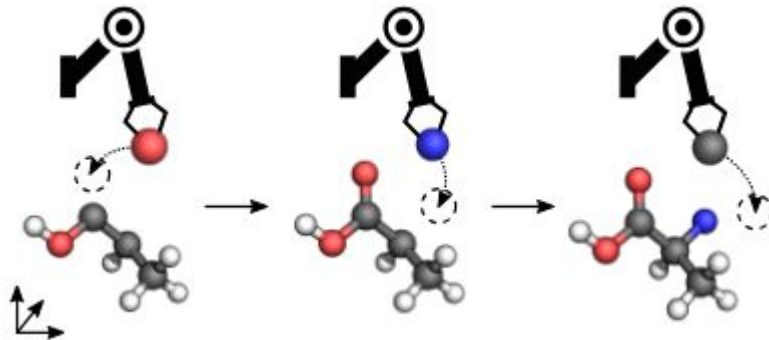
- We initialize $r(s)$ by counting points of chess pieces (pawn = 1, knight = 3, etc.) including *full look ahead of N moves*
 - e.g. $r = \max$ points we can win in 5 moves from this move
- We set two agents to play one another and adjust r based on the *actual* point differential from s'
- We include terms in the loss for winning and losing the game later on
- We update the policy (remember, those probabilities for a), typically with PPO

State of the art of RL in 2020



Simm, G.; Pinsler, R.; Hernandez-Lobato, J. M., *PMLR*, **2020**, 8959
Follow-up work by Simm et al.

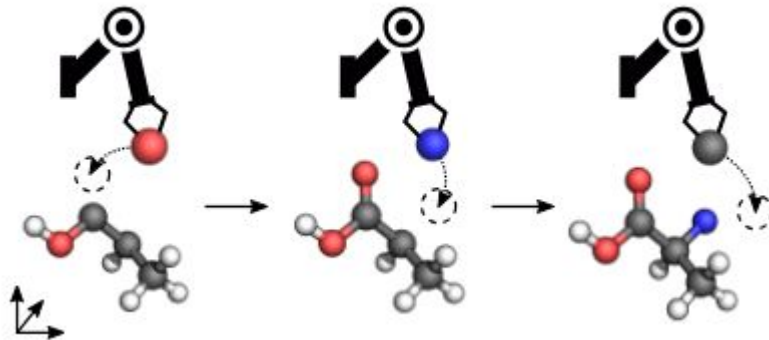
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- Actor-critic model in which we place atoms and see how energy changes (using quantum chemistry) as a reward to avoid unstable molecules
- Generates stable molecules, but very expensive *and* challenging.

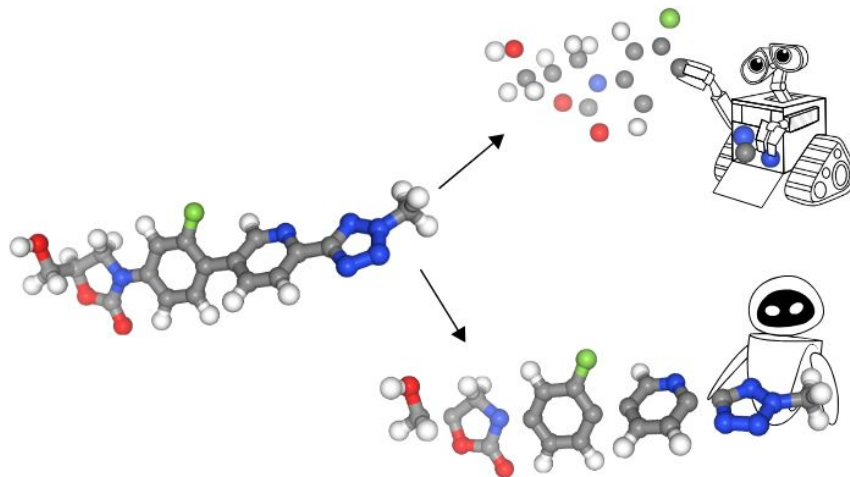
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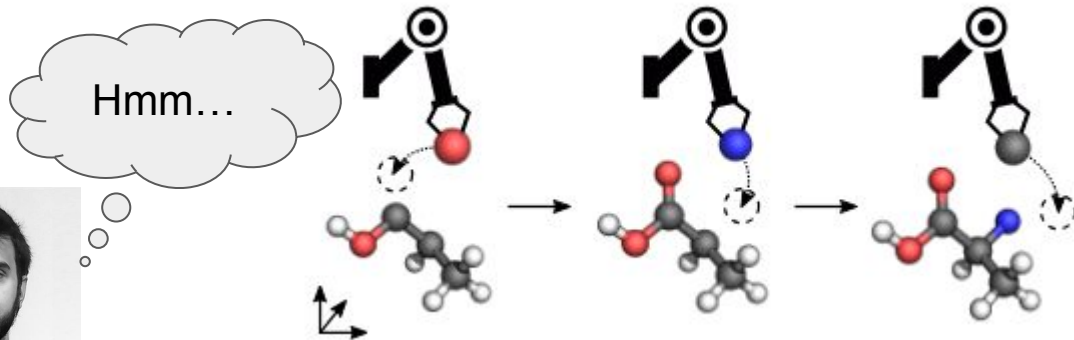
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Fiam-Shepherd, D.; Zhigalin, A.; Aspuru-Guzik, A. *arXiv*, **2022**

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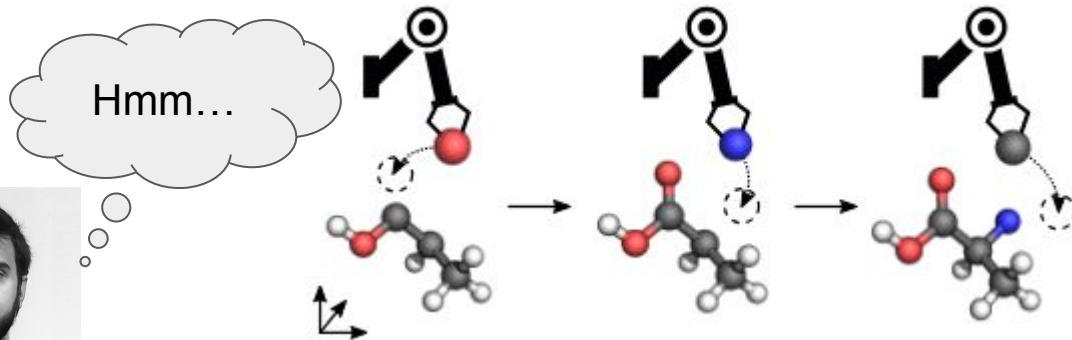
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Follow-up work by Simm et al.

- Is building a good catalyst a Markov chain?
 - If PPh_3 is good, is placing carbons on a P good? Why? **How?**

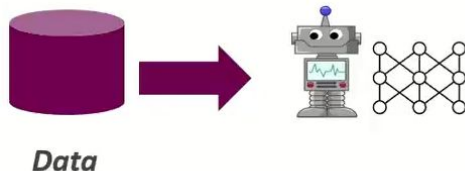
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Simm, G.; Pinsler, R.; Hernandez-Lobato, J. M., *PMLR*, **2020**, 8959
Follow-up work by Simm et al.

- Is building a good catalyst a Markov chain?
 - If PPh₃ is good, is placing carbons on a P good? Why? How?
 - Also, why do we care? Why can't we just put PPh₃ instead of building it atom by atom?

State of the art of molecular design in 2020-2024



Generative model that has learned the SMILES syntax

Olivecrona, M.; Blaschke, T.; Engkvist, O.; Chen, H. *J. Cheminform.* **2017**, 9 (1), 48

Blaschke, T.; Arús-Pous, J.; Chen, H.; Margreitter, C.; Tyrchan, C.; Engkvist, O.; Papadopoulos, K.; Patronov, A. REINVENT 2.0: An AI Tool for De Novo Drug Design. *J. Chem. Inf. Model.* **2020**, 60 (12), 5918

Loeffler, H. H.; He, J.; Tibo, A.; Janet, J. P.; Voronov, A.; Mervin, L. H.; Engkvist, O. Reinvent 4: Modern AI-Driven Generative Molecule Design. *J. Cheminform.* **2024**, 16 (1), 20

State of the art of molecular design in 2020-2024

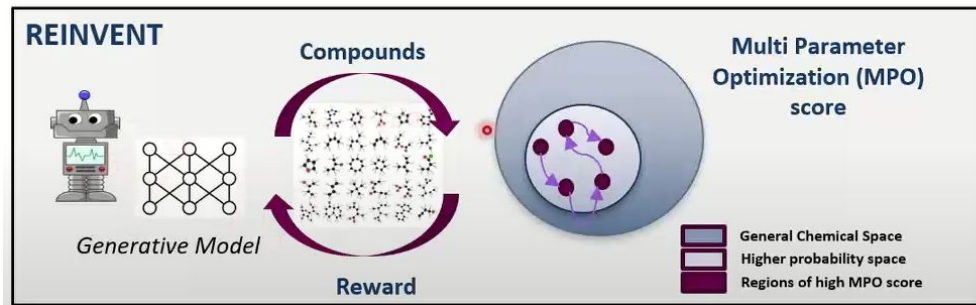


Data

Generative model that has learned the SMILES syntax

The RL is used to tune the **sampling!**

Seems obvious, but this is smart!



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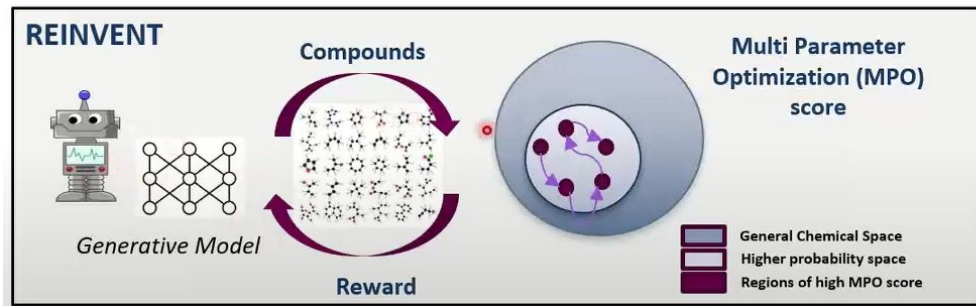
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State of the art of molecular design in 2020-2024



Flexible approach to
combine PPO with almost
any generative model



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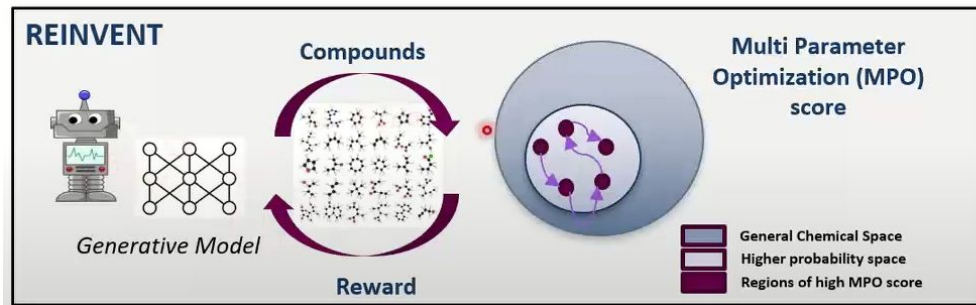


Data

Generative model that has learned the SMILES syntax

Flexible approach to combine PPO with almost any generative model

Expensive to condition (bottlenecked by eval. and inference speed)

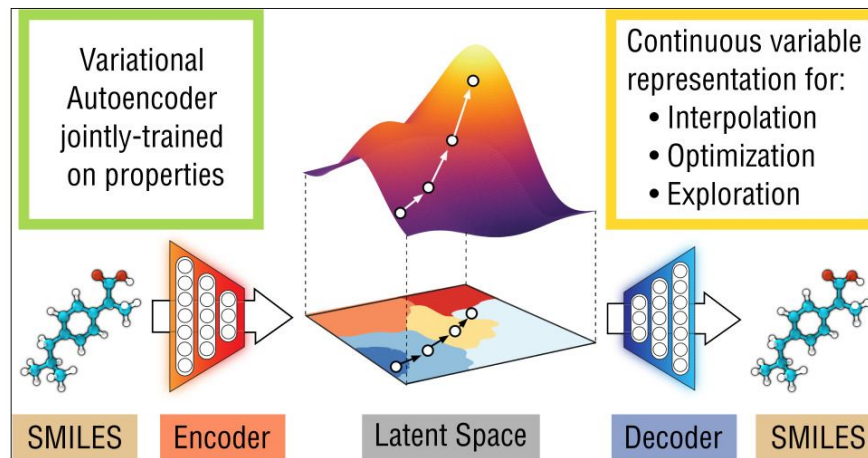


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State of the art of molecular design in 2020



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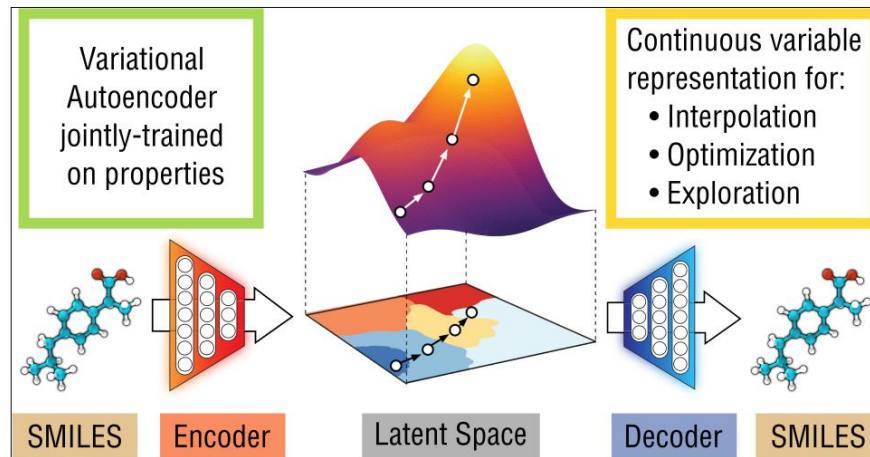
Krenn, M.; Häse, F.; Nigam, A.; Friederich, P.; Aspuru-Guzik, A. *Mach. Learn.: Sci. Technol.* **2020**, 1 (4), 045024

State of the art of molecular design in 2020: VAEs



Cool **but** works on
SMILES/SELFIES

- SMILES is bad for metals.
- SELFIES cannot deal with metals, even if it might be better.



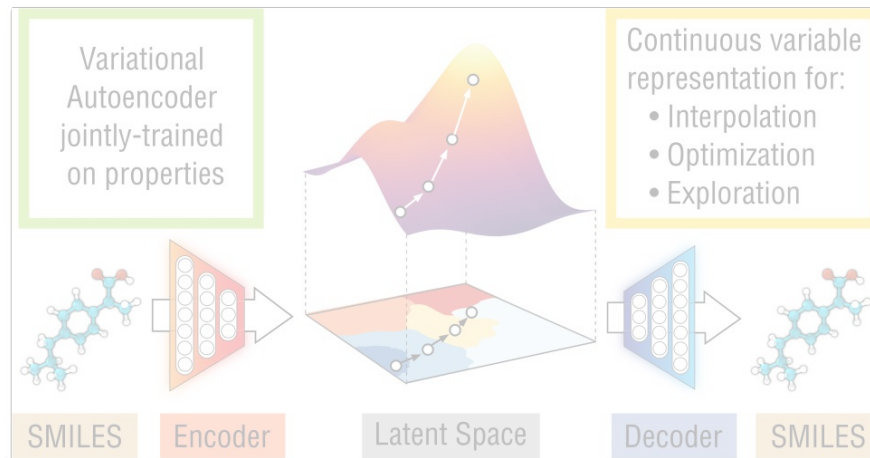
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State of the art of molecular design in 2020: VAEs



Cool **but** works on
SMILES/SELFIES

- SMILES is bad for metals.
- SELFIES cannot deal with metals, even if it might be better.
- **The joint training is hard**
- Would need lots of data and good architecture



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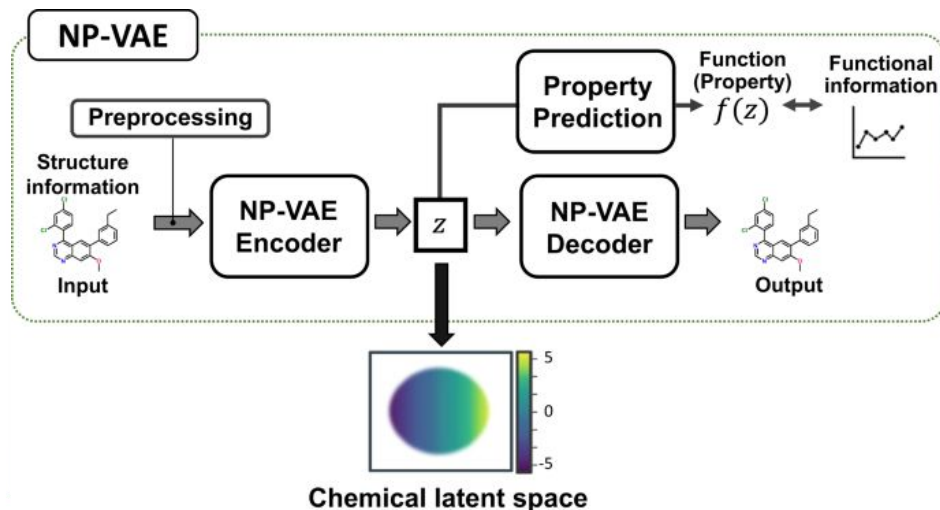
State of the art of molecular design in 2020: VAEs



How discrete is the decoding?

Decoding is not perfect!
It's trained on a continuous to discrete transformation.

We may decode wrongly!



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Dollar, O.; Joshi, N.; Pfaendtner, J.; Beck, D. A. C. *J. Phys. Chem. A* **2023**, 127 (37), 7844

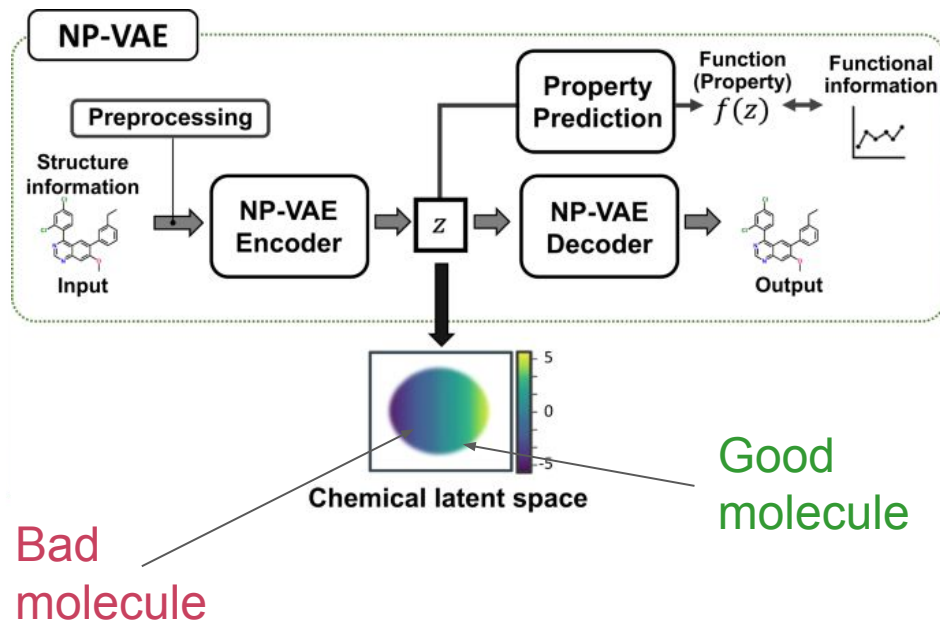
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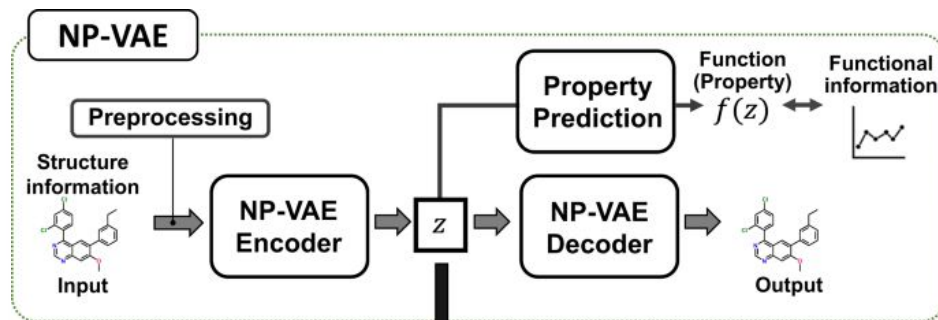
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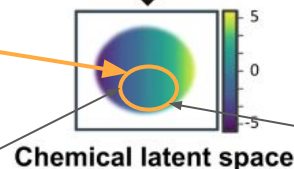
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Whole area decodes to the same molecule

Bad molecule



Good molecule

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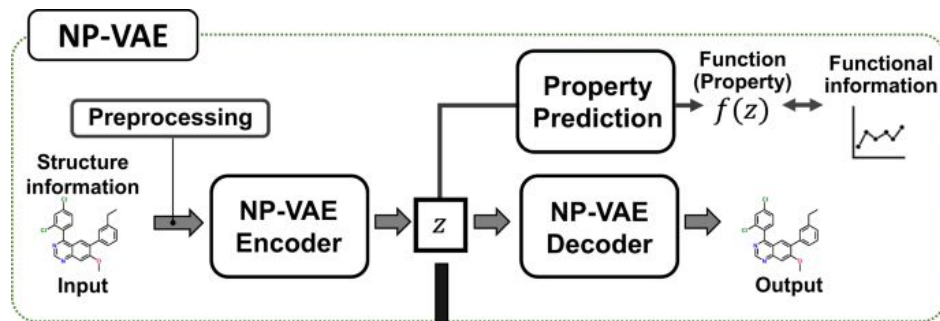
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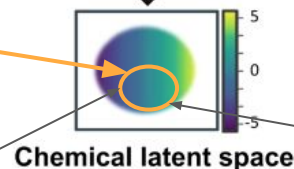
How discrete is the decoding?

Decoding is not perfect!
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We may decode wrongly!
Need a lot of data to smooth



Whole area decodes to the same molecule



Oopsie!

Whoopsie!

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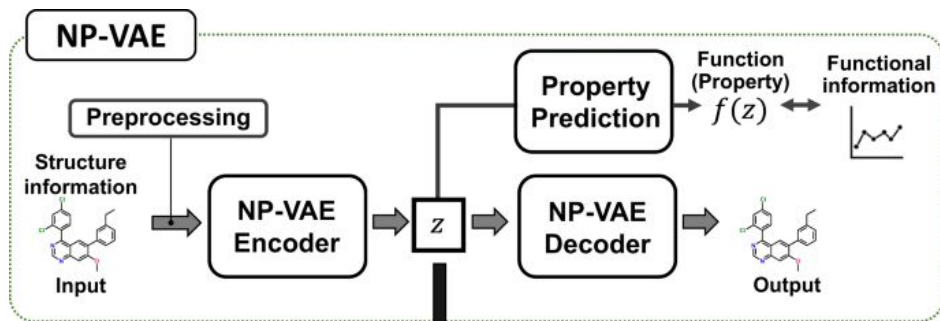
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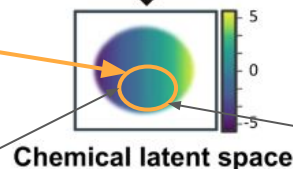
How discrete is the decoding?

Decoding is not perfect!
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We may decode wrongly!
Need a lot of data to smooth
Most efficient optimization



Whole area
decodes to the
same molecule



Oopsie!

Whoopsie!

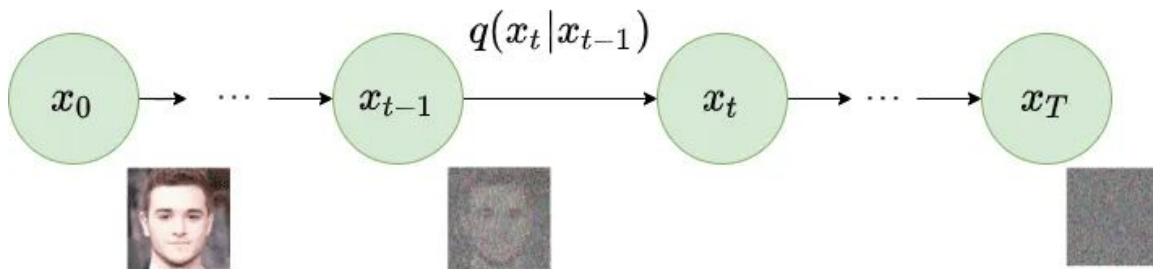
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State of the art of molecular design in 2022-2024: GDM

Newer approach

Goal: model a denoising process

We start by taking samples and adding noise in steps (driven by a scheduler)

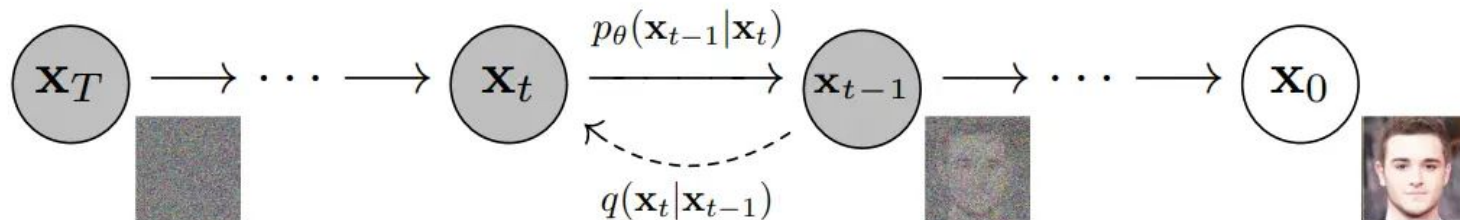


State of the art of molecular design in 2022-2024: GDM

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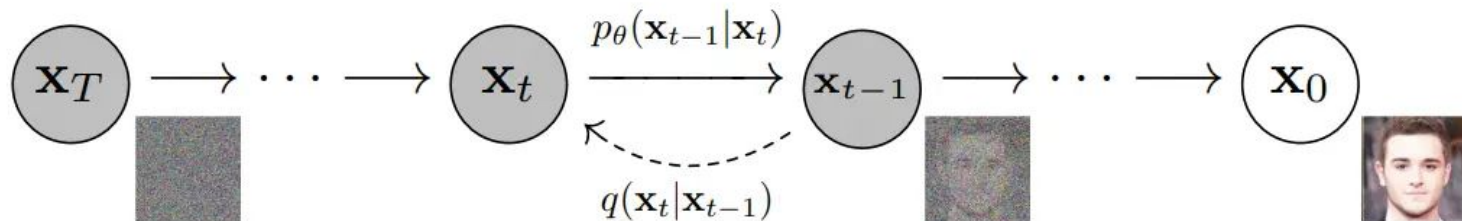
We train (basically a NN) to “go” from each step to the original
That is, the role of the NN is to **identify all the noise**

State of the art of molecular design in 2022-2024: GDM

Newer approach

Goal: model a denoising process

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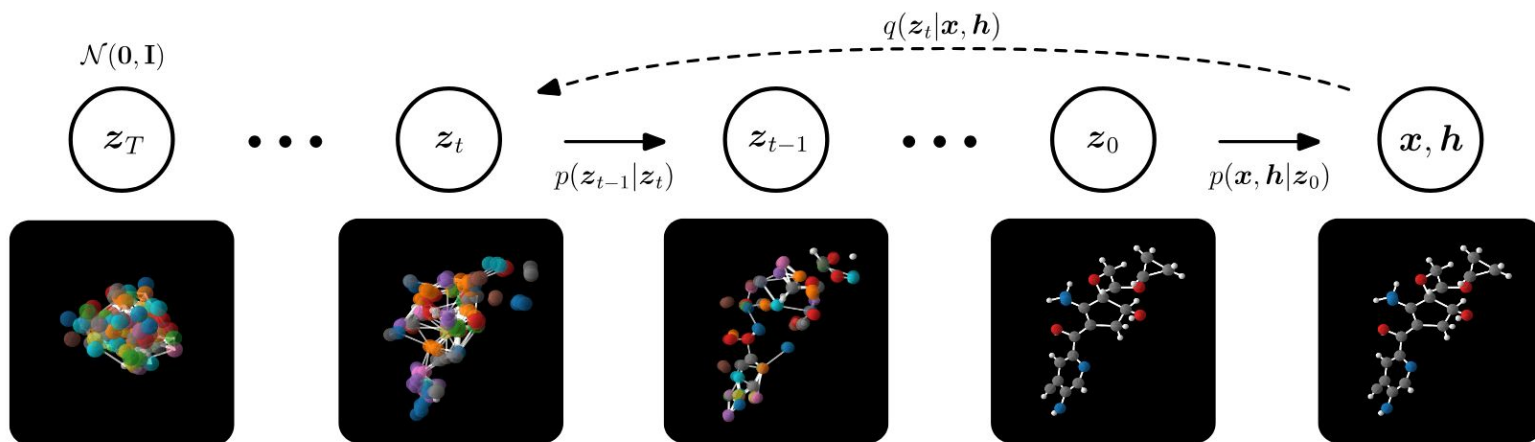
We train (basically a NN) to “go” from each step to the original

That is, the role of the NN is to identify all the noise

Still, we subtract a part of it (driven by a scheduler again) and repeat for...
some number of iterations

State of the art of molecular design in 2022-2024: GDM

We start by taking 3D structures and adding noise in atomic labels and positions
In principle compatible with all atom types but critically dependent on features

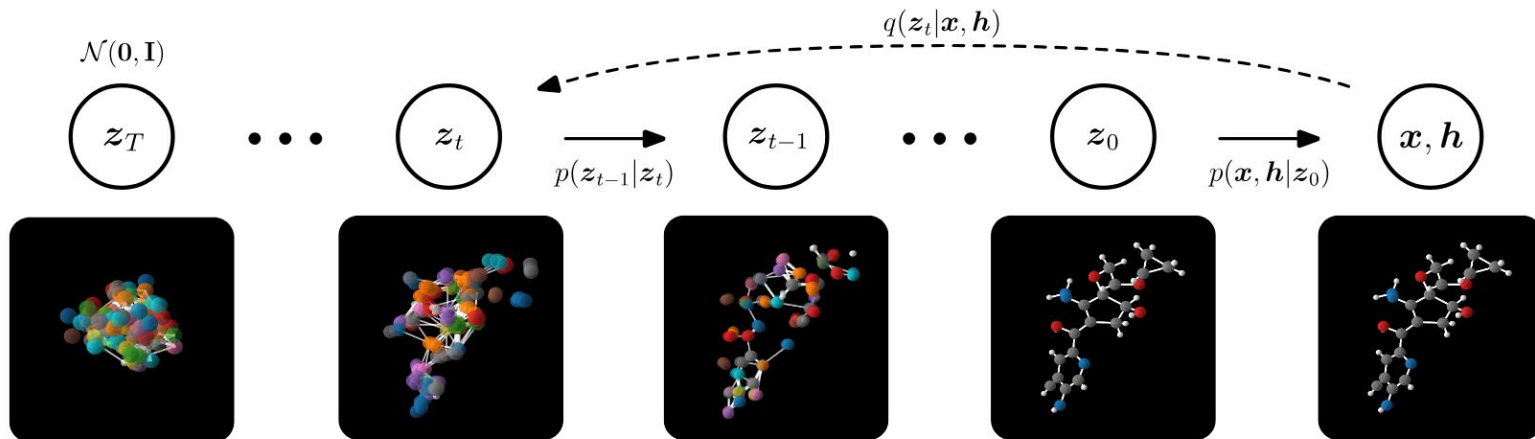


Hoogeboom, E.; Satorras, V. G.; Vignac, C.; Welling, M. PMLR, **2022**

State of the art of molecular design in 2022-2024: GDM

We include a separate regression model (that works on 3D coords. for instance) and use it to **bias the denoising** towards good properties

This is less certain than gradient optimization in VAEs, but almost as flexible as RL



Hoogeboom, E.; Satorras, V. G.; Vignac, C.; Welling, M. PMLR, **2022**

State of the art of molecular design in 2022-2024: GDM



So many moving parts!

Even taking most of the architecture for granted (i.e. we can define noise and gradients for features and coordinates equivariantly), still need:

- **Biasing is hard and relies on a separate model**
- **To define our schedulers and their rates (critical hyperparam.)**
- **To define feature sets that represent atom types**
- **Sampling is expensive (need to denoise N steps)**

State of the art of molecular design in 2022-2024: GDM



So many moving parts!

Even taking most of the architecture for granted (i.e. we can define noise and gradients for features and coordinates equivariantly), still need:

- Biasing is hard and relies on a separate model
- To define our schedulers and their rates (critical hyperparam.)
- To define feature sets that represent atom types
- Sampling is expensive (need to denoise N steps)
- On the other hand, **extremely powerful solution and easier to train**

Limitations towards fully automated inverse design

- Fitness determination still arguably the hardest challenge
- A good *fitness* solves a lot of problems (why would bad molecules have good fitness), specially in multiobjective settings
- Models that decouple generation and fitness evaluation are convenient but raise applicability domain concerns → active learning workflows
- Ensuring that the molecules are “**valid**” is an extremely difficult task
 - Defining “valid” as Lewis-like (e.g. SMILES-compliant) is too limiting
 - What does valid really mean? *Synthesizable? Stable?*



A comparative view

- Generative models (*note that I have skipped over flow models and RNNs and others*) are naturally stronger at interpolating to create truly new things
- GAs require stronger heuristics whereas generative models need more data
- Decoupling generation and property prediction is scalable but a price is paid in model applicability issues
 - VAEs are bottlenecked by training and novelty
 - Diffusion is bottlenecked by guidance complexity
 - RL approaches are bottlenecked by inference cost
 - GAs are “bottlenecked” by human-derived heuristics